

Recall words : vector space, subspace

Suppose $S \subset$ Vector space V (S may be infinite)

$$\text{Span}(S) := \left\{ \sum_{i=1}^n c_i v_i \mid c_i \in \mathbb{R}, v_i \in S \right\}$$

We say S is lin indep if $\left(\sum_{i=1}^n c_i v_i = 0 \text{ for } v_i \in S \right) \Rightarrow \begin{cases} c_i = 0 \\ \forall i \end{cases}$

We say S is a basis for V if $\begin{cases} S \text{ is lin indep} \\ \text{Span}(S) = V \end{cases}$

$\dim(V)$ = cardinality of any basis of V

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Need to prove : existence of basis & uniqueness of dim

Example : ① The following is a basis for $\mathbb{R}^n$

$\{e_1, \dots, e_n\}$ where $e_i = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \rightarrow i\text{th slot}$

$\sum c_i e_i = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$. Easy to see that $\{e_i\}_{i=1, \dots, n}$ is a basis

Other bases for $\mathbb{R}^n$? Yes!

$\mathbb{R}^1$: any nonzero elt is a basis

$\mathbb{R}^2$: $\{u, v\}$ where u, v (nonzero) vectors not lying on the same line thro' the origin

$\mathbb{R}^3$: any 3 vectors that do not lie in a plane thro' the origin

② A Basis for poly of deg $\leq n$ is $\{1, x, x^2, \dots, x^n\}$

$c_0 + c_1 x + c_2 x^2 + \dots + c_n x^n$ is "sort of like" vector

$\begin{bmatrix} c_0 \\ \vdots \\ c_n \end{bmatrix}$ in $\mathbb{R}^{n+1}$ [We will make this precise when we define the notion of an isomorphism]

· A basis for all poly $\{1, x, x^2, x^3, \dots\} = \{x^n \mid n=0, 1, \dots\}$

This is an infinite dim vector space.

· Similar for poly in multiple variables

$$f_A: \mathbb{R}^C \rightarrow \mathbb{R}^r$$

③ Given $r \times c$ matrix A & associated lin map $x \mapsto Ax$

how to find basis for $\{ \text{nul}(A) \text{ i.e. } \ker(f_A) \}$

$\{ \text{col}(A) \text{ i.e. } \text{range}(f_A) \}$

Do this using $\text{RREF}(A)$

Burning questions to answer: Let V be a vspce

Q1. Does V have a basis? YES

Q2. Do any two bases have the same cardinality? YES

Imp case $V = \mathbb{R}^n$

Let $v_1, v_2, \dots, v_k \in \mathbb{R}^n$. Then we saw

① $\text{Span} \{v_1, \dots, v_k\} = \mathbb{R}^n \Rightarrow k \geq n$

② $v_1, \dots, v_k$ lin indep $\Rightarrow k \leq n$

$\{v_1, \dots, v_k\}$ is a basis $\Rightarrow k=n$

Reason for ① & ② Consider $n \times k$ matrix A whose columns are $v_1, \dots, v_k$!

① $\Leftrightarrow Ax=b$ has a soln for each b

$\Leftrightarrow \text{RREF}(A)$ must have a pivot in each row

$\Rightarrow \# \text{rows} = n \leq \# \text{cols} = k$

② $\Leftrightarrow Ax=0$ has only the trivial soln

$\Leftrightarrow \text{RREF}(A)$ must have a pivot in each column

$\Rightarrow k \leq n$

Observe: n vectors in $\mathbb{R}^n$ span $\mathbb{R}^n \Leftrightarrow$ they are lin indep

What about Q1 & Q2 for general v.space V ?

How to find a basis? Two strategies

① "Start with empty set and keep adding indep vectors till they span"

Observe suppose $S \subset V$, S indep, $\text{span}(S) \neq V$

Then $S \cup \{u\}$ is independent

Pf $\sum_{v \in S} \alpha_v v + \lambda u = 0 \Rightarrow \begin{cases} \text{If } \lambda = 0, \text{ all } \alpha_v = 0 \text{ by indep of } S \\ \text{If } \lambda \neq 0, u \in \text{span}(S) \text{ contradiction} \end{cases}$

problem: termination?

② "Start with a spanning set & keep deleting vectors till remaining part is indep"

Observe $S \subset V$ Suppose $v \in S$ and $v \in \text{span}(S - \{v\})$
Then $\text{span}(S) = \text{span}(S - \{v\})$

[Notation for sets A, B $A \setminus B = \{a \in A \mid a \notin B\}$]

problem Termination, unless I happen to start with a "good" spanning set

For now we'll restrict to vspaces that have a finite spanning set. Then strategy B is guaranteed to work! (A is strategy A, as we will see)
 Done in next lecture

Defⁿ V is called a finite dimensional vector space (fdvsp) if V has a finite spanning set.

Thm ① Every fdvsp V has a basis

② Any two bases have the same cardinality

Pf ① ✓ by strategy B

② We will give two proofs of this tomorrow

— concrete using $\mathbb{R}^n$ and notion of isomorphism

— more abstract using Steinitz exchange lemma

Added after class

Looking ahead to tomorrow: try the following

- Read the definition of isomorphism. Isomorphic vector spaces "obviously" have the same dimension. Show that any fdim v.space V is isomorphic to $\mathbb{R}^n$ (for some n). Deduce ② for V .

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This will be our concrete proof of ②. For abstract proof observe the following easy lemma

- Lemma (Steinitz exchange property): Suppose $x \in S \subset V$ and $u \in V$ such that $u \in \text{Span}(S)$, $u \notin \text{Span}(S - \{x\})$. Then $x \in \text{Span}(S - \{x\} \cup \{u\})$. [Note: we have exchanged u for x ; hence the name]

- Use the lemma to show the following in any v-space V
let $I = \text{a finite linearly indep set}$ } Then $|I| \leq |S|$ (*)
 $S = \text{any spanning set}$

- This easily finishes the abstract proof of ② for f.dim v-spaces:

- We also need to do the following for f.dim v-spaces:
 - proof that any indep set I can be enlarged to get a basis, so strategy ① works to get a basis.

In class I first said we'll deduce this using (*). But in fact it is much simpler. Just take any finite Spanning set S (which exists b/c V is f-dimensional) and add vectors from S to I as needed to get a basis. We did this simple demonstration of strategy A in the next class.